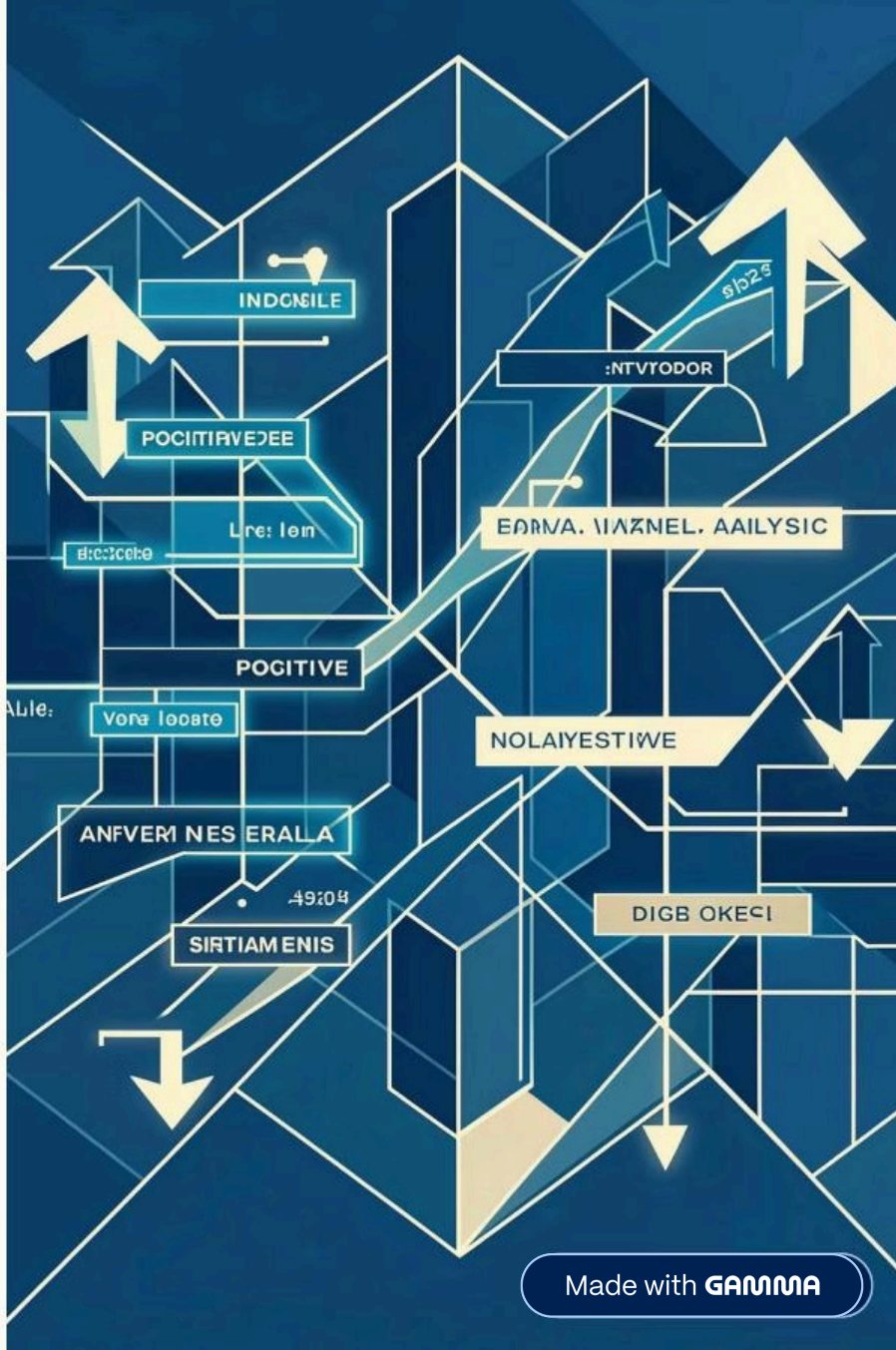
Sentiment Analysis using Machine Learning

Darshana Chaudhari - Data Analytics Internship



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Problem Statement

1

Analyze Text Reviews

Extract meaningful insights from unstructured text data.

2

Classify Sentiment

Categorize reviews as Positive or Negative.

3

Use NLP Techniques

Apply Natural Language Processing for text understanding.

4

Build NLP Model

Develop a robust classification model.

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Dataset Overview

50,0...

Total Records

Extensive text-based review data.

—

Sentiment Classes

Positive & Negative for binary classification.

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Balanced Dataset

Ensuring fair representation for training.

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Data Preprocessing Steps

01

Converted to Lowercase

Standardizing text for consistent analysis.

02

Removed Punctuation & Stopwords

Cleaning text to focus on meaningful words.

03

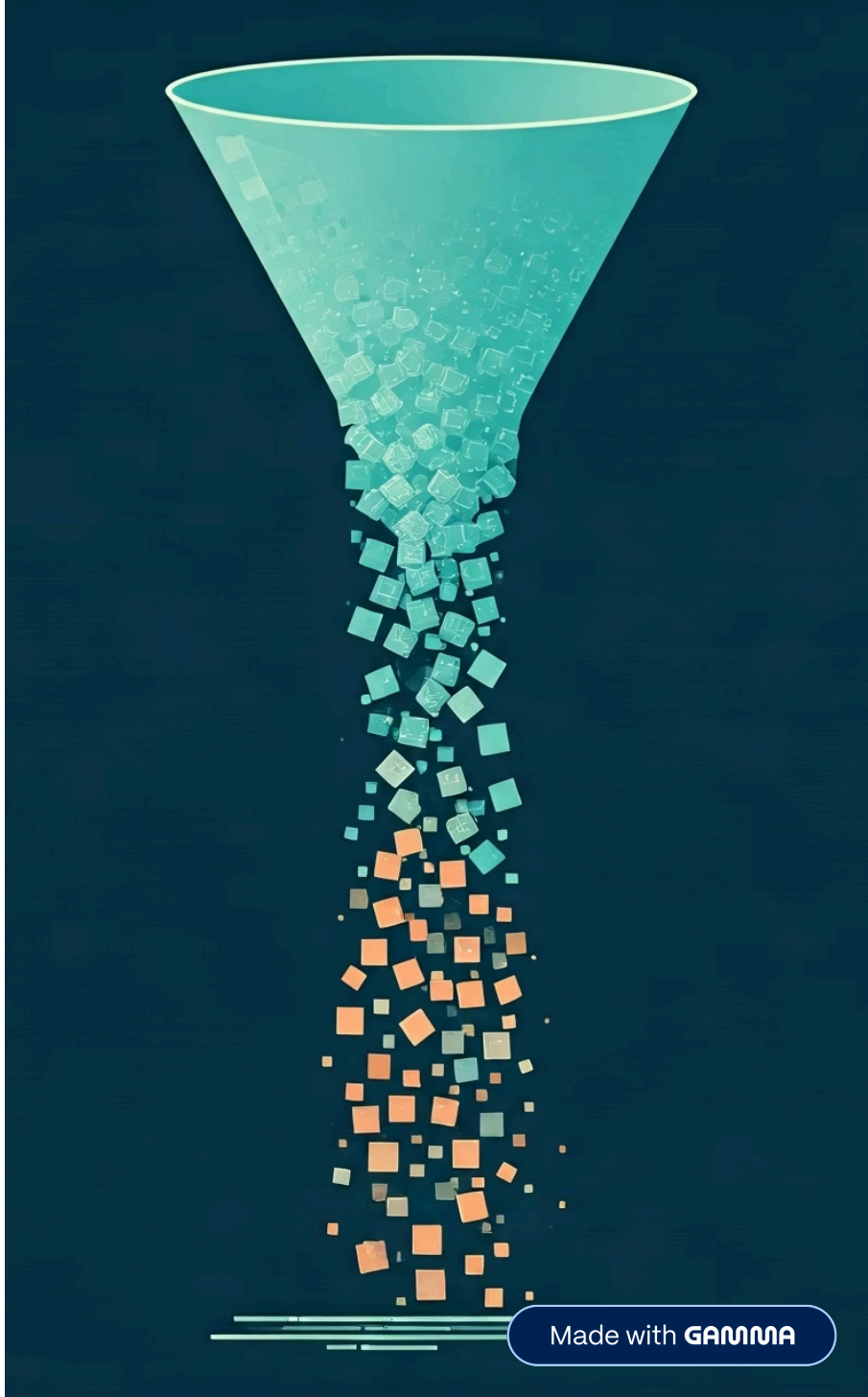
Tokenization

Breaking text into individual words or phrases.

04

Applied TF-IDF Vectorization

Converting text into numerical features for the model.



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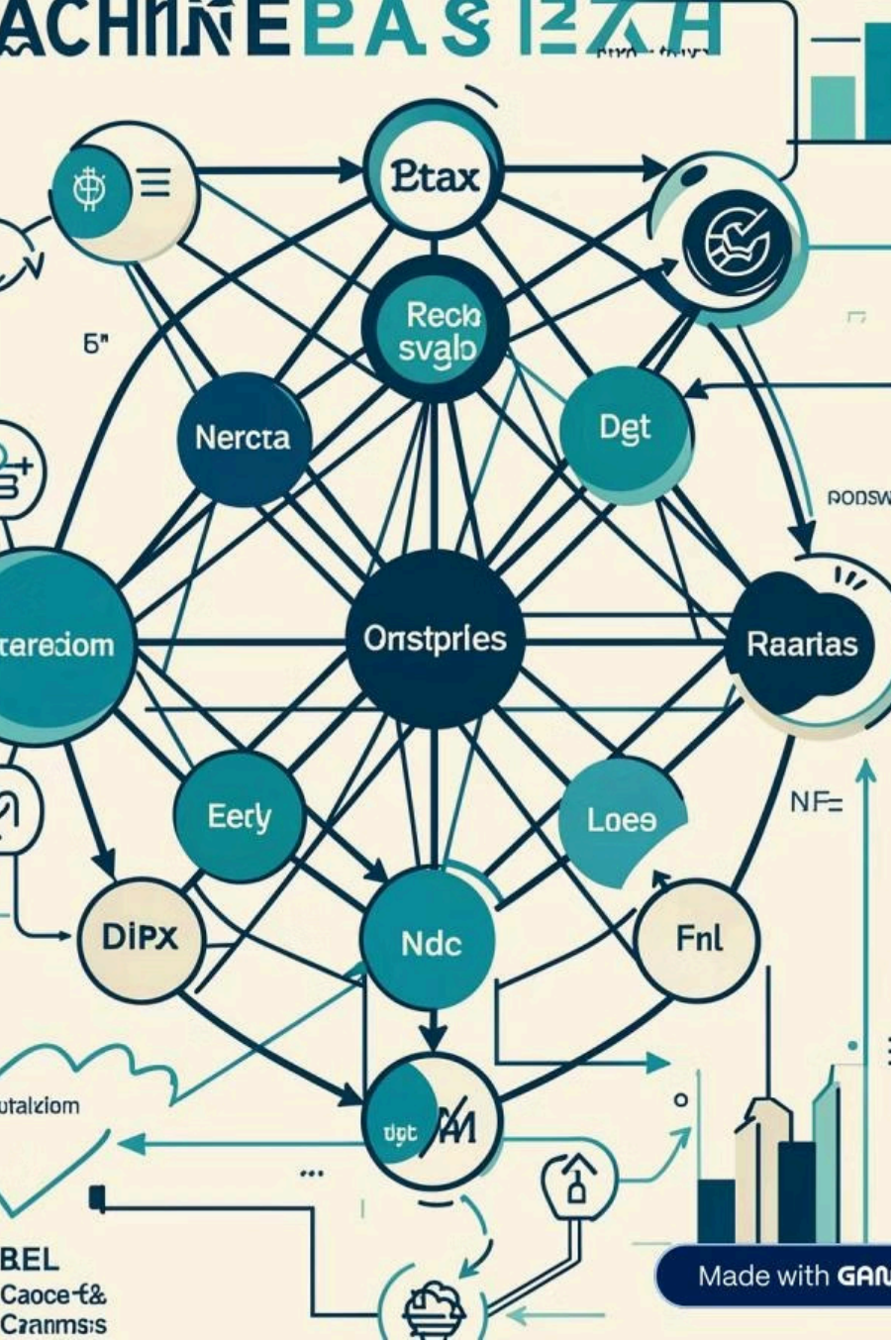

Model Used: Multinomial Naive Bayes

Supervised Learning

Trained on labeled data.

Binary Classification

Distinguishes between two classes.

Train-Test Split (80:20)

Ensuring robust model evaluation.

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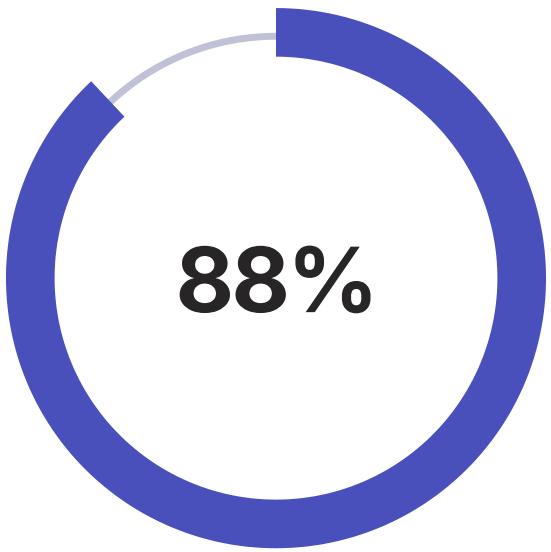
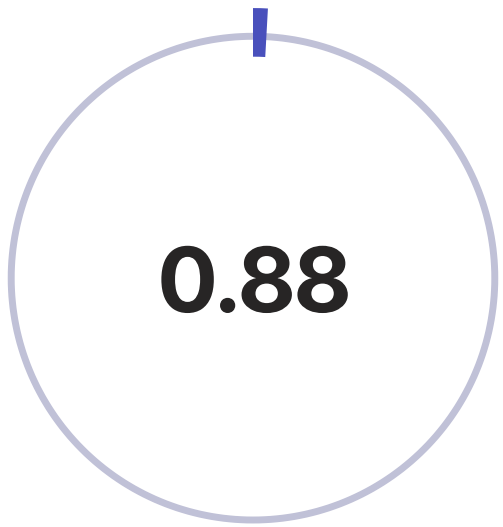
Model Performance



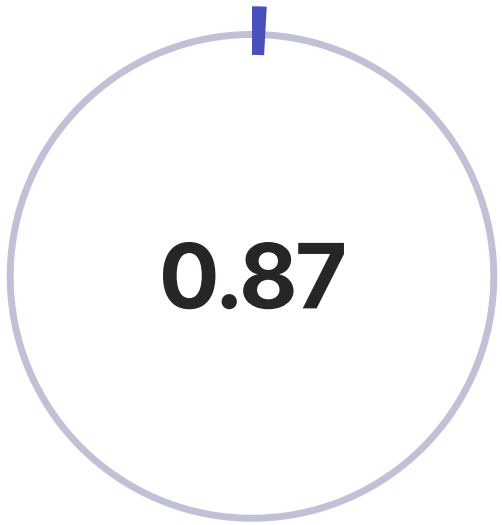
Accuracy

Overall correctness of predictions.



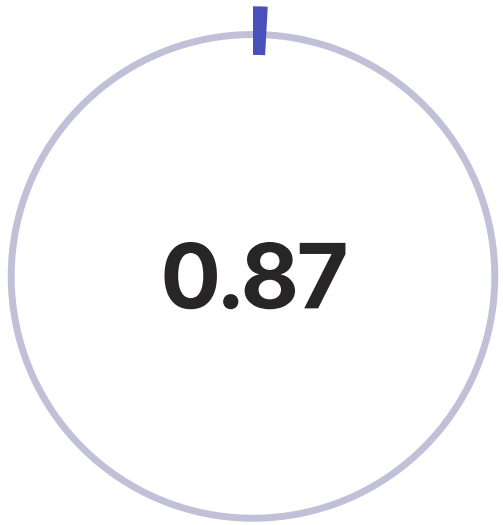
Precision

Proportion of true positive predictions.



Recall

Ability to identify all relevant instances.



F1 Score

Harmonic mean of precision and recall.

> The model demonstrates balanced and effective performance across key metrics.

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Key Insights & Applications

Key Insights

- Model performs well on balanced data.
- Positive reviews slightly easier to classify.
- Text preprocessing improved accuracy.
- TF-IDF effective for feature extraction.

Applications

Customer Feedback Analysis

Social Media Monitoring

Brand Reputation Management

Market Research

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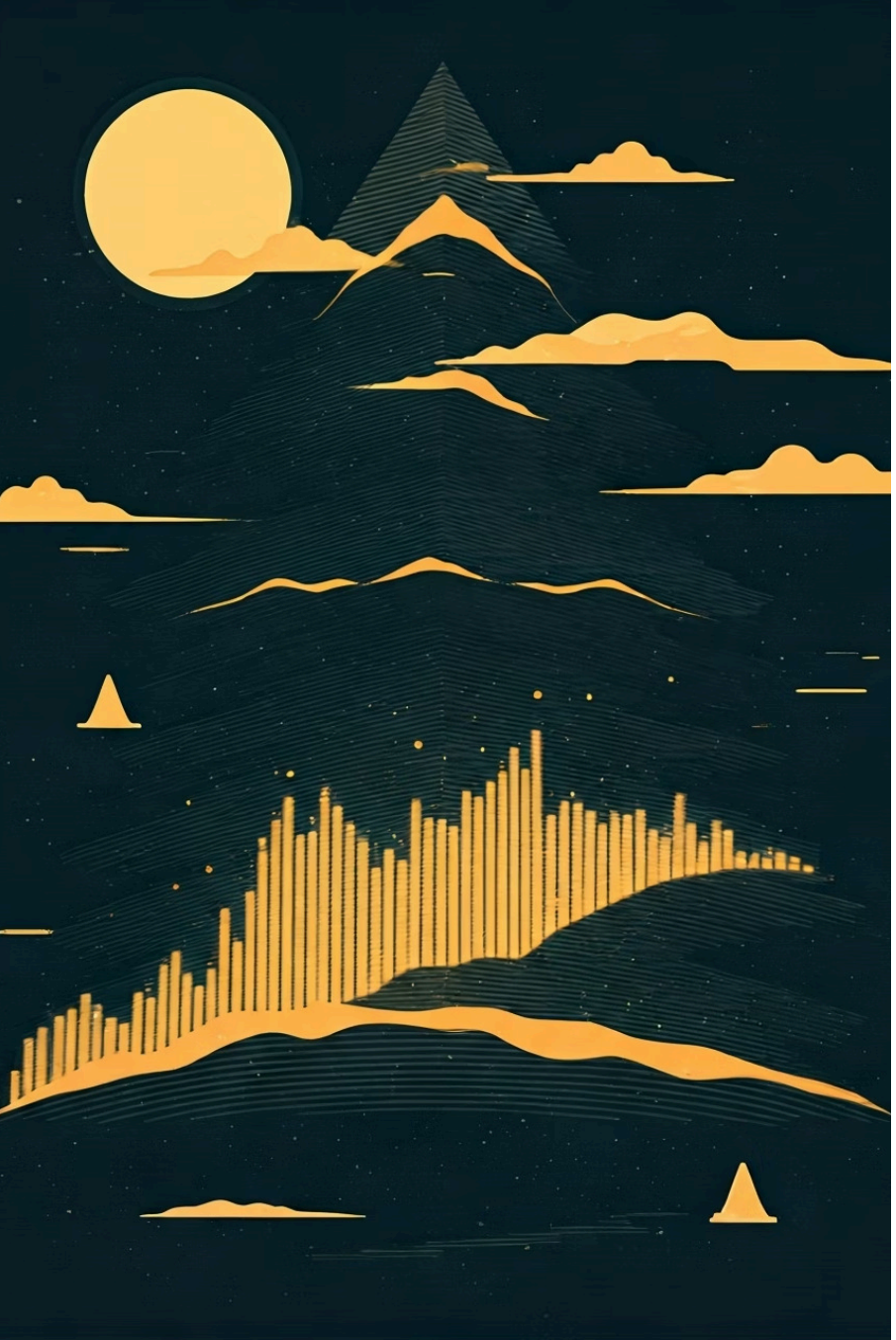
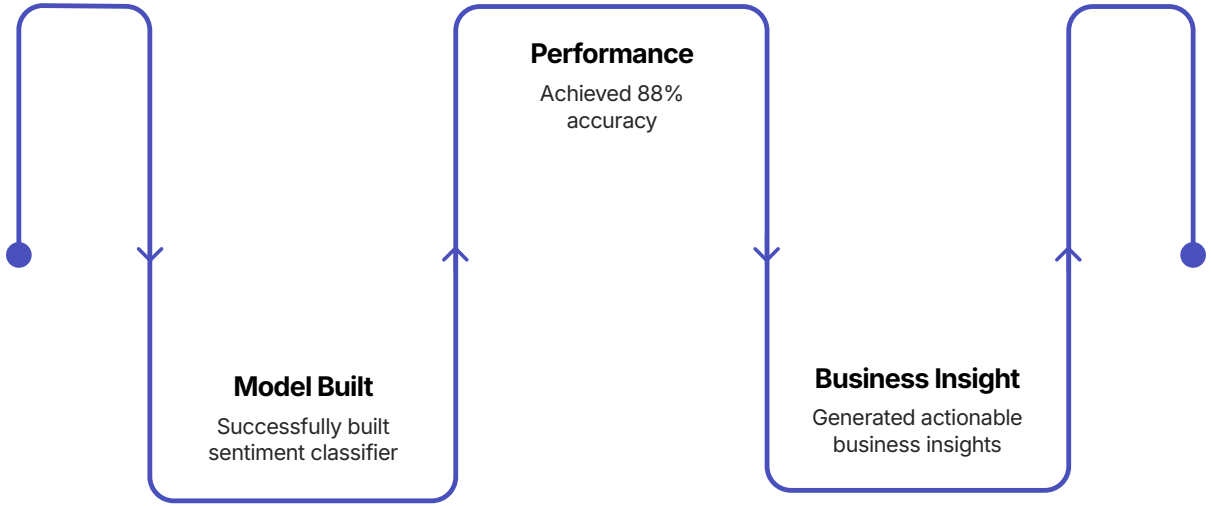


Conclusion



This project successfully demonstrated the power of NLP and machine learning in extracting valuable sentiment from text data, providing a foundation for various business applications.

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